

VU DINH BAO PHUC

Software Engineer (Erlang/BEAM) | CKAD & PSM I Certified

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ABOUT ME

Software Engineer with 4 years of professional experience specializing in Erlang/BEAM architecture, concurrent programming, and high-availability distributed systems within the telecommunications domain. Proven track record of managing core IMS infrastructure handling massive-scale traffic, engineering distributed states (ETS/Mnesia), and directly ownership-handling critical production incidents (24/7 HOT tickets) under tight SLAs. As a certified CKAD and PSM I professional, I am proficient in cloud-native microservices (Kubernetes, Docker, Helm), automated CI/CD pipelines, and driving Agile/Scrum team efficiency. Highly passionate about leveraging cutting-edge AI/LLM tools to optimize engineering workflows and drive system optimization.

Career Interests: Eager to apply my distributed systems expertise, functional programming mastery, and analytical problem-solving skills to tackle complex scalability challenges in diverse, cutting-edge engineering domains.

EDUCATION

University of Information Technologies | 2019 - 2023

Engineer in Computer Engineering

- GPA: 8.15
 - Full scholarship 2019 - 2020
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WORK EXPERIENCE

Endava Company | Jun 2022 - Now

Software Engineer

Project Description: Developing a P-CSCF and IBCF in the IP Multimedia Subsystem (IMS) which is commonly used by many operators telecommunication network all over the world. Our product is supporting multimedia in IMS application such as Voice over LTE or 5G, Wi-Fi-Calling and many interconnect scenarios.

Work Contributes:

- Core Erlang & OTP Architecture: Designed, refactored, and maintained high-availability backend components utilizing OTP primitives, specifically engineering Gen Servers and managing concurrent Erlang processes to achieve strict fault isolation and system resilience.

- Core Functional Domain Logic: Programmed complex backend features across 4 critical telecom domains:
 - Signaling & Characteristic: Handled high-volume SIP signaling traffic and optimized Characteristic Domain data structures to dynamically adapt IMS components to diverse operator network environments.
 - Fault Management (FM) & Logging: Implemented real-time fault isolation, logging behaviors, and automated alert rules directly within the Erlang codebase.
 - Performance Management (PM): Engineered backend telemetry tracking to collect, aggregate, and expose critical system performance metrics.
- Cloud-Native Infrastructure:
 - Containerization: Containerized Erlang-based services using Docker and managed deployments on Kubernetes to enhance system self-healing and scalability.
 - Data storage: Integrated Redis from Valkey for high-performance data caching and storage within microservices.
 - Observability: Leveraged Fluent Bit to build centralized logging pipelines, ensuring seamless log streaming and traceability across Kubernetes clusters.
 - System Configuration: Implemented and integrated YANG Provider to handle complex system configurations, ensuring standardized data modeling and device management.
- BEAM Storage & Throughput Optimization: Resolved severe system performance bottlenecks by optimizing complex database operations (read, match, and delete) within native BEAM storage layers (ETS and Mnesia tables).
- System Testing & Reliability: Executed a rigorous 3-phase testing strategy (Unit, Functional, and Node-level system testing) on SUSE Linux environments to guarantee the stability of core telecom services.
- Global Stakeholder Collaboration: Actively collaborate with stakeholders and cross-functional teams across Canada, USA, Hungary, Sweden, and China to maintain, support and answer question from global product deployments.
- Critical Incident Response (24/7): Member of the core 24/7 technical team, acting as the primary owner for "HOT" tickets (Critical/Urgent Incidents) and directly debugged live distributed Erlang nodes and deployed runtime hot-fixes under tight SLA pressure for global operators.

Scrum Master

Project Description: Acted as a facilitator for a cross-functional team, promoting Agile values and ensuring seamless project execution through the removal of operational bottlenecks and alignment with business goals.

Work Contributes:

- Strategic Problem Solving: Collaborated with other Scrum Masters and Team Coaches to conduct Root Cause Analysis (RCA) for complex systemic issues, proposing and implementing cross-team solutions to improve overall organizational efficiency.
- Balanced Workflow Management: Successfully managed technical debt and maintenance tasks using Kanban alongside Scrum for feature development, ensuring a strategic balance between urgent production fixes and long-term milestones.

- Stakeholder Bridging: Acted as a vital link between leadership directives and technical execution, translating high-level business objectives into actionable, well-defined technical tasks.
- Conflict & Dependency Resolution: Proactively identified and resolved internal team conflicts and external dependencies, shielding the development team from distractions and maintaining focus on high-priority deliverables.

Line Manager Internship Program

- **Project Description:** Served as a technical mentor and lead for the internship program, bridging the gap between academic knowledge and industrial production standards through hands-on guidance and project oversight.
- **Work contributes:**
 - Technical Mentorship: Guided interns through the full software development lifecycle (SDLC), providing deep dives into company-standard technologies including Erlang and Docker.
 - Quality Oversight: Conducted rigorous Code Reviews and technical evaluations, ensuring interns' contributions met production-grade quality, security, and performance standards.
 - Environment Provisioning: Managed and deployed project environments, including Kazoo server deployment and documentation server setup, to provide interns with a realistic development sandbox.
 - Performance Coaching: Delivered structured feedback at the end of each sprint, helping interns track their growth and improve their technical proficiency throughout the program.

University of Information and Technologies | 2019 - 2023

Measuring device and LoRaWAN gateway's coverage display system

- **Project Description:** This project involved researching and developing a device to measure signal strength for LoRaWAN gateways and a system to display the coverage area of LoRaWAN gateways.
- **Work contributes:**
 - Designed a GPS-enabled device capable of determining location and measuring signal strength between the device and the LoRa gateway on Altium Designer and Arduino IDE.
 - Developed a server to act as a mediator between The Things Network cloud and Firebase Database, ensuring seamless data transmission.
 - Created an application to visualize the signal strength between the device and the LoRa gateway through an interactive heatmap based on React Native Framework.

SKILLS

- **Programming Languages:**
 - Erlang (Advanced): OTP, ETS, Mnesia, GenServer
 - Bash Script (Intermediate)
 - JavaScript, Python (Basic)
- **Tools and Framework:** YANG, Docker, Kubernetes, React JS, K9S, Prometheus, Grafana
- **Platforms and Cloud technologies:** Linux, VS Code, Gitlab, Jenkins, Gerrit, Fluent Bit, Redis
- **Methodologies and other tools:** Agile, Scrum, Kanban, Jira, Confluence, CKAD, PSM I
- **AI Tools:** Kiro, AmazonQ, Windsurf, Miro, Github Copilot